ORBCOMM®

WHERE DATA DRIVES DECISIONS




# DB 300 (Distress Button) Installation Guide

GJD139, Version 01
Aug 2025

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# TABLE OF CONTENTS


Contact and Legal Information ..... 2
Contact Support ..... 2
Trademark Notice ..... 2
Preface ..... 4
Purpose ..... 4
WEEE Statement ..... 4
Battery Safety Warnings ..... 4
General Warnings ..... 4
1 Installation ..... 5
1.1 Tools and Materials ..... 5
1.2 Install the Device ..... 5
1.3 Power the Device ON ..... 6
1.4 Pair the DB 300 with the Asset ..... 6
1.5 Signal Distress ..... 6
1.6 Cancel a Distress Alert ..... 7
1.7 Replace the Batteries ..... 7
APPENDIX A Hardware Specifications ..... 8
APPENDIX B LED Functionality ..... 9
APPENDIX C Inclement Weather Guidelines ..... 10



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# PREFACE

## Purpose

This guide contains product information for the DB 300 (wireless distress button). The intended audiences for this guide include field support personnel, product evaluators, and certified third-party personnel. It is particularly intended for personnel who are responsible for system installation and activation. In addition, and as is appropriate, this guide may be used for training purposes.

## WEEE Statement

Disposal of this product should be handled according to all national laws and regulations.

| | |
|---|---|
| The mark shown to the right is in compliance with Waste Electrical and Electronic Equipment Directive 2002/96/EC (WEEE). The mark indicates the requirement NOT to dispose of the equipment as unsorted municipal waste but use the return and collection systems according to local law. |  |

## Battery Safety Warnings

CAUTION: Always follow local disposal guidelines to properly dispose of the Lithium-ion battery and the device.

CAUTION: Store in a cool, well-ventilated area. Elevated temperatures will result in shortened battery life.

## General Warnings

**IMPORTANT**

READ ALL INSTRUCTIONS CAREFULLY BEFORE INSTALLING. FAILURE TO DO SO MAY CAUSE PERSONAL INJURY OR DAMAGE TO PRODUCT AND/OR PROPERTY.

• Review the product package and contents prior to beginning the installation. Take care when opening the packaging and removing items. If a return is needed, you will want to return the product in its original packaging if possible. • This instruction guide is provided as a GENERAL installation guide; some assets vary dimensionally and may require additional steps. • ORBCOMM has a policy of continuous development and improvements. Therefore, products, guides, and technical information are subject to change without prior notice. • The manufacturer and / or distributors do not accept responsibility for third-party charges, labor, and / or third-party replacement modifications that are not ORBCOMM approved. Some modifications may void the factory warranty. • ORBCOMM does not accept any responsibility for installations performed by installers / third parties not approved and / or authorized by ORBCOMM. Some installations may void the factory warranty. • Exercise due diligence when installing this product. ORBCOMM does not accept any responsibility for asset damage or personal injury resulting from the installation of this product. Careless installation and operation can result in serious injury or equipment damage. • All liability for installation and use rests with the owner / operator. • Always make sure you have a clean, dry, and well-lit work area. • Always ensure products are secure during disassembly and installation. • Always take steps to protect yourself when drilling, cutting, and grinding because this may create flying particles that can cause injury. • Thoroughly inspect the area to be drilled, on both sides of material, prior to modification, and relocate any objects that may become damaged. • Always route electrical cables carefully. Avoid moving parts, parts that may become hot and rough, or sharp edges. • Make sure to fully understand the product, its intended use, and operation prior to use.

CAUTION: While ORBCOMM provides mounting hardware to assist with installations, it is the responsibility of the installer to select the proper mounting hardware for the asset's surface material where an ORBCOMM device or accessory will be mounted.


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# 1 INSTALLATION

The DB 300 is a wireless distress button designed to report critical alerts in remote and off-grid environments. With ultra-long battery life and long-range Bluetooth, it enables reliable connection with compatible devices. The DB 300 is ideal for safety-critical applications where immediate alerting is essential, and connectivity is limited.



## 1.1 Tools and Materials

The DB 300 shipping box includes the following:

- DB 300
- Bracket



- Double-sided tape
- Screw and rubber plug

Customer supplied materials:

- Cable ties

## 1.2 Install the Device

The DB 300 offers several installation options: screw, cable ties, or double-sided tape. Use the method that best suits your installation requirements.

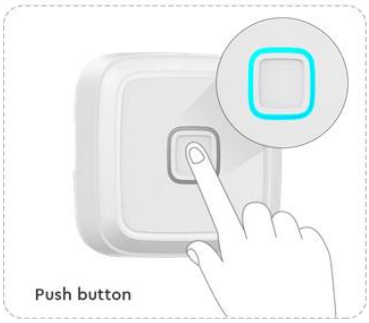


Screws Cable Ties Double-sided Tape

Note: If using the double-sided tape method, refer to APPENDIX C.


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## 1.3 Power the Device ON




Power ON: Long button press - press and hold the button down for 3 seconds, and the multicolor LED flashs green for 3 seconds to indicate device power is ON.

## 1.4 Pair the DB 300 with the Asset

Refer to *[MA002] Field Support Tool (FST) User Guide* to pair the DB 300 (Distress Button) with the terminal.

## 1.5 Signal Distress

1. Press and hold the distress button for 3 seconds to activate an alert.
    - The button's LED begins flashing RED, indicating that the DB 300 is transmitting the distress alert over the satellite network.

      
    - Once the alert is successfully sent, the LED changes to flashing BLUE, confirming that the distress message was successfully transmitted.

      
    - When the monitoring platform (Solution Provider's back-end application) acknowledges the alert, the LED


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changes to flashing GREEN to indicate that the alert has been received and confirmed.



Note: The distress state remains active until it is deactivated.

Note: While active, the button does not accept additional activations.

## 1.6 Cancel a Distress Alert

Once a distress signal has been addressed, it is possible to cancel the distress alert over-the-air to reset the DB 300 so that it can report new distress alerts.

For details on how to cancel a distress alert, please refer the *[GJD141] SC 1000 Reference Guide (for Partners)*.

## 1.7 Replace the Batteries

The DB 300 requires four ER14505 batteries.




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# APPENDIX A HARDWARE SPECIFICATIONS

| Item | Specifications | |
|---|---|---|
| Casing | Dimensions | 7.5.cm x 7.5 cm x 2.5 cm (3" x 3" x 1") |
| | Material | PC+ASA |
| | Waterproof | IP67 |
| | Color | White |
| | Installation | Double-sided tape / bracket (screw or cable zip tie) |
| Connectivity | Bluetooth | BLE 5.1 |
| | Long range mode[1] | Yes |
| | Transmission range[2] | Legacy mode – 150 meters (165 yards)<br>Long range mode – 350 meters (380 yards) |
| Hardware | LED | Annular RED indicator (RGB optional) |
| | Capacity | 10400 mAh |
| | Replaceable battery | Yes |
| | Storage temperature | -20°C to +70°C / -4°F to +158°F (without battery)<br>10°C to + 25°C / 50°F to +77°F (with ER14505 battery) |
| | Operating temperature | (with ER14505 battery) -20°C to +60°C / -4°F to +140°F<br>Use of this product in adverse conditions will void the product warranty. |
| Regulatory | Certifications | FCC \| CE \| RoHS \| REACH |

[1]Transmission range tested in the open area and no obstacles in the route.

[2]Long range mode means “Coded PHY advertising mode”.


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# APPENDIX B LED FUNCTIONALITY

| LED State | Meaning |
|---|---|
| OFF | Button is inactive (no active alert) |
| Flashing RED | Transmitting distress alert over satellite |
| Flashing BLUE | Alert successfully transmitted over satellite |
| Flashing GREEN | Alert acknowledged by the monitoring platform (Solution Provider's back-end) |


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# APPENDIX C INCLEMENT WEATHER GUIDELINES

**Mandatory Guidelines for Installation in Wet Weather**

The installation surface on the asset must be completely clean and dry for the tape to bond. If it is raining or snowing hard enough that the surface cannot be kept dry, DO NOT proceed with the installation.

**Mandatory Guidelines for Installation in Cold Weather**

Below 15°C (60°F) the double-sided tape starts becoming firm which makes it more difficult to bond to the asset.

If the guidelines below are followed correctly, the device can be installed at temperatures down to -20°C (-5°F).

- At or below freezing temperatures (0°C or 32°F), both the device and the tape primer* must be kept at room temperature, for example, inside an idling vehicle or a warm building.

  *If tape primer is not provided with the device, the cold temperature installation kit (ORBCOMM p/n ST101505, suitable for up to 25 installs) MUST be used.

Example of Tape Primer




- Keep the primer warm (room temperature) until ready to apply. The primer will not dry quickly at cold temperatures however, in this situation the tape should be applied when the primer is still wet, as it improves initial bond.
- Keep the device warm (room temperature) until it is time to mount it to the asset.
- Press firmly on the entire top surface of the device (7 kg (15 lb) for 15 seconds) to bond the tape to the asset.

Failing to follow these guidelines will compromise the installation.


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